LAB ASSIGNMENT 1
COMPUTER NETWORKS

-PARITOSH LAHRE(12341550)

Q1)

```
❯ ifconfig
br-ad240cce97c7: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.18.0.1  netmask 255.255.0.0  broadcast 172.18.255.255
        ether 5e:5c:d2:84:e6:60  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether ca:dd:a2:71:9d:c7  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 39 overruns 0  carrier 0  collisions 0

eno2: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        ether 58:11:22:41:c8:bd  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 16  memory 0x60100000-60120000

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 6702  bytes 1236345 (1.2 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 6702  bytes 1236345 (1.2 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

virbr0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 192.168.122.1  netmask 255.255.255.0  broadcast 192.168.122.255
        ether 52:54:00:bb:bd:3c  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 49 overruns 0  carrier 0  collisions 0

wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.50.19.54  netmask 255.255.192.0  broadcast 10.50.63.255
        inet6 fe80::5191:2a04:3cb3:9ae2  prefixlen 64  scopeid 0x20<link>
        ether 70:a6:cc:1b:74:c5  txqueuelen 1000  (Ethernet)
        RX packets 1346692  bytes 1686889461 (1.6 GB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 432940  bytes 71622324 (71.6 MB)
        TX errors 0  dropped 17 overruns 0  carrier 0  collisions 0
```

Command Executed: `ifconfig`

Output Description: The command lists all network interfaces on the system (including virtual bridges like `docker0` and the loopback `lo`). The active network interface connecting the system to the Local Area Network is **wlo1** (Wireless LAN).

a)

Important Attributes of wlo1:

- **inet (IPv4 Address):** `10.50.19.54`
  - This is the logical address currently assigned to the machine on the network.
- **ether (Hardware/MAC Address):** `70:a6:cc:1b:74:c5`
  - This is the unique physical identifier for the network interface card.
- **netmask:** `255.255.192.0`
  - This mask defines the range of the local subnet.
- **RX packets:** `1346692` (approx. 1.6 GB)
  - "RX" stands for Received. This indicates the total amount of data downloaded by this interface since the last reset.
- **TX packets:** `432940` (approx. 71.6 MB)
  - "TX" stands for Transmitted. This indicates the total amount of data uploaded or sent by this interface.
- **UP, RUNNING:**
  - These flags indicate that the driver is loaded and the interface is actively communicating with the network.

b)

1. `-a`: Displays all interfaces, including those that are down.
2. `up` / `down`: Activates or deactivates an interface (e.g., `ifconfig eth0 down`).
3. `promisc`: Puts the interface into promiscuous mode (sniffing all traffic).
4. `mtu N`: Sets the Maximum Transmission Unit to N bytes.

Q2)

a) The network statistics (netstat) command is a networking tool used for troubleshooting and configuration that can also serve as a monitoring tool for connections over the network. Both incoming and outgoing connections, routing tables, port listening, and usage statistics are common uses for this command.

b)

```
❯ sudo netstat -tap
[sudo] password for paritosh:
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 paritosh-ASUS-TU:domain 0.0.0.0:*               LISTEN      2108/dnsmasq
tcp        0      0 _localdnsstub:domain    0.0.0.0:*               LISTEN      1064/systemd-resolv
tcp        0      0 _localdnsproxy:domain   0.0.0.0:*               LISTEN      1064/systemd-resolv
tcp        0      0 localhost:61209         0.0.0.0:*               LISTEN      1955/python3
tcp        0      0 localhost:mysql         0.0.0.0:*               LISTEN      2031/mysqld
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN      1954/cupsd
tcp        0      0 localhost:33060         0.0.0.0:*               LISTEN      2031/mysqld
tcp        0      0 paritosh-ASUS-TUF:48624 a23-63-216-40.dep:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:46618 104.18.31.173:https     ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:36848 server-18-66-41-1:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:45582 server-18-66-41-1:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:37352 pnbomb-ad-in-f3.1:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:37074 104.16.174.226:https    ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:59844 a23-32-39-144.dep:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:42914 server-13-227-249:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:49736 server-13-227-249:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:37344 pnbomb-ad-in-f3.1:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:45952 a23-218-249-185.d:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:53876 server-18-66-57-2:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:45576 server-18-66-41-1:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:44216 server-18-172-78-:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:52134 server-18-172-78-:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:50674 54.30.211.130.bc.:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:59570 47.164.160.34.bc.:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:49738 server-13-227-249:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:57226 136.255.227.35.bc:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:35094 pnbomb-bp-in-f14.:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:35770 server-13-227-249:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:57954 sd-in-f84.1e100.n:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:35762 server-13-227-249:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:34054 a23-63-221-112.de:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:51956 server-18-66-57-5:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:42056 151.101.2.137:https     ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:55148 a23-218-249-179.d:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:59202 53.16.211.130.bc.:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:60872 server-18-239-153:https ESTABLISHED 4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:34046 a23-63-221-112.de:https CLOSE_WAIT  4617/renderD129 --c
tcp        0      0 paritosh-ASUS-TUF:57134 bom07s26-in-f14.1:https ESTABLISHED 4617/renderD129 --c
tcp6       0      0 ip6-localhost:ipp       [::]:*                  LISTEN      1954/cupsd
```

Command Executed: `sudo netstat -tap`

Analysis of Output:

- **Active TCP Ports:** The output shows several services listening on local TCP ports:
  - **DNS (Port 53):** Identified as `domain` in the output. PIDs **2108** (`dnsmasq`) and **1064** (`systemd-resolv`) are listening.
  - **MySQL (Port 3306):** Identified as `mysql`. PID **2031** (`mysqld`) is listening.
  - **IPP (Port 631):** Identified as `ipp` (Internet Printing Protocol). PID **1954** (`cupsd`) is listening.
  - **Python:** A python process (PID **1955**) is listening on port `61209`.
- **Web Browser Identification:**
  - **PID: 4617**
  - **Process Name:** In the screenshot, the process connecting to external **HTTPS** servers (Foreign Address ending in `:https`) is listed with PID **4617**. (Note: The name appears as `renderD129` or truncated, which is likely a

sub-process of the browser or an Electron-based application managing the web rendering).
    - **Explanation:** I identified this as the browser because it has multiple `ESTABLISHED` connections to remote servers on port 443 (HTTPS).
- **Can you identify the port number and PID of a specific TAB? No.** As seen in the output, all the HTTPS connections share the same PID (**4617**). Modern browsers use a multi-process architecture but often route network traffic through a specific parent or network process. `netstat` shows the socket belonging to that process ID, making it impossible to distinguish between different tabs just by looking at this output.
- **Services using standard ports:**
    - **HTTP (80):** Not found listening.
    - **DNS (53): Found.** Used by `dnsmasq` (PID 2108) and `systemd-resolv` (PID 1064).
    - **DHCP (67/68):** Not found in TCP list (DHCP usually uses UDP).
    - **SMTP (25) / FTP (21):** Not found.

c)

```
❯ netstat -su
IcmpMsg:
    InType0: 171
    InType3: 460
    InType8: 19
    OutType0: 19
    OutType3: 548
    OutType8: 187
Udp:
    1351101 packets received
    560 packets to unknown port received
    31623 packet receive errors
    319666 packets sent
    31623 receive buffer errors
    147 send buffer errors
UdpLite:
IpExt:
    InNoRoutes: 7
    InMcastPkts: 457
    OutMcastPkts: 1448
    InOctets: 1897257622
    OutOctets: 67337796
    InMcastOctets: 59564
    OutMcastOctets: 253667
    InNoECTPkts: 1465956
    InECT1Pkts: 49038
    InECT0Pkts: 2997
    InCEPkts: 1
MPTcpExt:
```

Command Executed: `netstat -su`

Output Explanation: The option **`-s`** displays summary statistics for each protocol, and **`-u`** filters this specifically for **UDP**.

- **Analysis:** The output shows that the system has received **1,351,101** UDP packets and sent **319,666** UDP packets. It also reports **31,623** packet receive errors, indicating some network traffic issues or buffer overflows.

Q3)

a) Ping (Packet Internet Groper) is a basic Internet command used in computer network administration to test and verify the availability of a destination IP address. It helps

users confirm whether the computer they are trying to reach is operational. Ping works by measuring the communication latency which is the time taken for data to travel between two devices or across a network. As communication latency decreases, the effectiveness of communication improves and hence makes it an essential tool for network troubleshooting. It can be used with any operating system (OS) that supports networking.

b)

i)

```
❯ ping -c 10 youtube.com
PING youtube.com (142.250.205.206) 56(84) bytes of data.
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=1 ttl=114 time=27.2 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=2 ttl=114 time=23.3 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=3 ttl=114 time=21.1 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=4 ttl=114 time=33.0 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=5 ttl=114 time=31.4 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=6 ttl=114 time=34.6 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=7 ttl=114 time=230 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=9 ttl=114 time=69.6 ms
64 bytes from sin11s28-in-f14.1e100.net (142.250.205.206): icmp_seq=10 ttl=114 time=20.8 ms

--- youtube.com ping statistics ---
10 packets transmitted, 9 received, 10% packet loss, time 9047ms
rtt min/avg/max/mdev = 20.783/54.567/230.089/63.611 ms
```

```
❯ ping -c 10 youtube.com
PING youtube.com (142.250.195.46) 56(84) bytes of data.
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=1 ttl=113 time=100 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=2 ttl=113 time=82.6 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=3 ttl=113 time=88.3 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=4 ttl=113 time=76.0 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=5 ttl=113 time=87.7 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=6 ttl=113 time=81.6 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=7 ttl=113 time=86.0 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=8 ttl=113 time=83.0 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=9 ttl=113 time=84.9 ms
64 bytes from maa03s37-in-f14.1e100.net (142.250.195.46): icmp_seq=10 ttl=113 time=104 ms

--- youtube.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
rtt min/avg/max/mdev = 76.032/87.385/103.609/7.999 ms
```

```
❯ ping -c 10 youtube.com
PING youtube.com (142.250.205.206) 56(84) bytes of data.
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=1 ttl=114 time=80.6 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=2 ttl=114 time=90.8 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=3 ttl=114 time=82.8 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=4 ttl=114 time=84.5 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=5 ttl=114 time=84.3 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=6 ttl=114 time=69.6 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=7 ttl=114 time=84.9 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=8 ttl=114 time=83.9 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=9 ttl=114 time=91.3 ms
64 bytes from lcboma-ba-in-f14.1e100.net (142.250.205.206): icmp_seq=10 ttl=114 time=82.5 ms

--- youtube.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
rtt min/avg/max/mdev = 69.602/83.518/91.304/5.660 ms
```

```
❯ ping -c 10 google.com
PING google.com (142.250.192.78) 56(84) bytes of data.
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=113 time=74.8 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=113 time=67.6 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=4 ttl=113 time=20.3 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=5 ttl=113 time=29.3 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=6 ttl=113 time=62.2 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=7 ttl=113 time=85.6 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=8 ttl=113 time=37.8 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=9 ttl=113 time=52.6 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=10 ttl=113 time=44.5 ms

--- google.com ping statistics ---
10 packets transmitted, 9 received, 10% packet loss, time 9024ms
rtt min/avg/max/mdev = 20.323/52.750/85.637/20.474 ms
```

```
❯ ping -c 10 google.com
PING google.com (142.250.192.78) 56(84) bytes of data.
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=1 ttl=113 time=83.7 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=113 time=73.2 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=113 time=64.3 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=4 ttl=113 time=21.4 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=5 ttl=113 time=49.2 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=6 ttl=113 time=76.6 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=7 ttl=113 time=68.8 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=8 ttl=113 time=73.8 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=9 ttl=113 time=61.3 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=10 ttl=113 time=80.2 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9007ms
rtt min/avg/max/mdev = 21.397/65.273/83.728/17.439 ms
```

```
❯ ping -c 10 google.com
PING google.com (142.250.192.78) 56(84) bytes of data.
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=1 ttl=113 time=46.8 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=113 time=64.9 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=113 time=80.6 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=4 ttl=113 time=77.0 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=5 ttl=113 time=74.8 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=6 ttl=113 time=65.0 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=7 ttl=113 time=91.1 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=8 ttl=113 time=86.7 ms
64 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=9 ttl=113 time=44.0 ms

--- google.com ping statistics ---
10 packets transmitted, 9 received, 10% packet loss, time 9012ms
rtt min/avg/max/mdev = 43.989/70.095/91.113/15.548 ms
```

```
❯ ping -c 10 instagram.com
PING instagram.com (57.144.242.34) 56(84) bytes of data.
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=1 ttl=52 time=22.6 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=2 ttl=52 time=56.3 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=3 ttl=52 time=28.0 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=4 ttl=52 time=33.5 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=5 ttl=52 time=59.8 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=6 ttl=52 time=20.5 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=7 ttl=52 time=18.1 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=8 ttl=52 time=47.0 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=9 ttl=52 time=20.3 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=10 ttl=52 time=35.0 ms

--- instagram.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9009ms
rtt min/avg/max/mdev = 18.137/34.108/59.787/14.568 ms
```

```
❯ ping -c 10 instagram.com
PING instagram.com (57.144.242.34) 56(84) bytes of data.
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=1 ttl=52 time=18.6 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=2 ttl=52 time=20.6 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=3 ttl=52 time=21.9 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=4 ttl=52 time=23.3 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=5 ttl=52 time=21.6 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=6 ttl=52 time=24.3 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=7 ttl=52 time=24.6 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=8 ttl=52 time=21.9 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=9 ttl=52 time=22.0 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=10 ttl=52 time=20.7 ms

--- instagram.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9011ms
rtt min/avg/max/mdev = 18.620/21.948/24.581/1.710 ms
```

```
❯ ping -c 10 instagram.com
PING instagram.com (57.144.242.34) 56(84) bytes of data.
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=1 ttl=52 time=80.7 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=2 ttl=52 time=78.3 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=5 ttl=52 time=87.5 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=6 ttl=52 time=82.3 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=7 ttl=52 time=89.2 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=8 ttl=52 time=90.7 ms
64 bytes from instagram-p42-shv-01-bom5.fbcdn.net (57.144.242.34): icmp_seq=10 ttl=52 time=72.7 ms

--- instagram.com ping statistics ---
10 packets transmitted, 7 received, 30% packet loss, time 9128ms
rtt min/avg/max/mdev = 72.651/83.061/90.660/6.014 ms
```

Average RTT Table: I observed the following Average Round Trip Times (RTT) across three different sessions:

Host	Session 1 (ms)	Session 2 (ms)	Session 3 (ms)	Server Location
instagram.com	34.108	21.948	83.061	Mumbai (bom5)
google.com	52.750	65.273	*N/A*	Mumbai (bom12)
youtube.com	54.567	87.385	83.518	Chennai (maa03)

Correlation Analysis:

- **Geographical Distance:** Yes, there is a correlation. The servers identified in **Mumbai** (closer to Bhilai) generally showed lower RTTs (approx. **21-65 ms**) compared to the server in **Chennai** (further away), which showed consistently higher RTTs (approx. **54-87 ms**).
- **Time of Day:** RTT fluctuated significantly between sessions (e.g., Instagram jumped from 21ms to 83ms), indicating that network traffic and congestion at different times of the day also heavily impact latency.

ii)

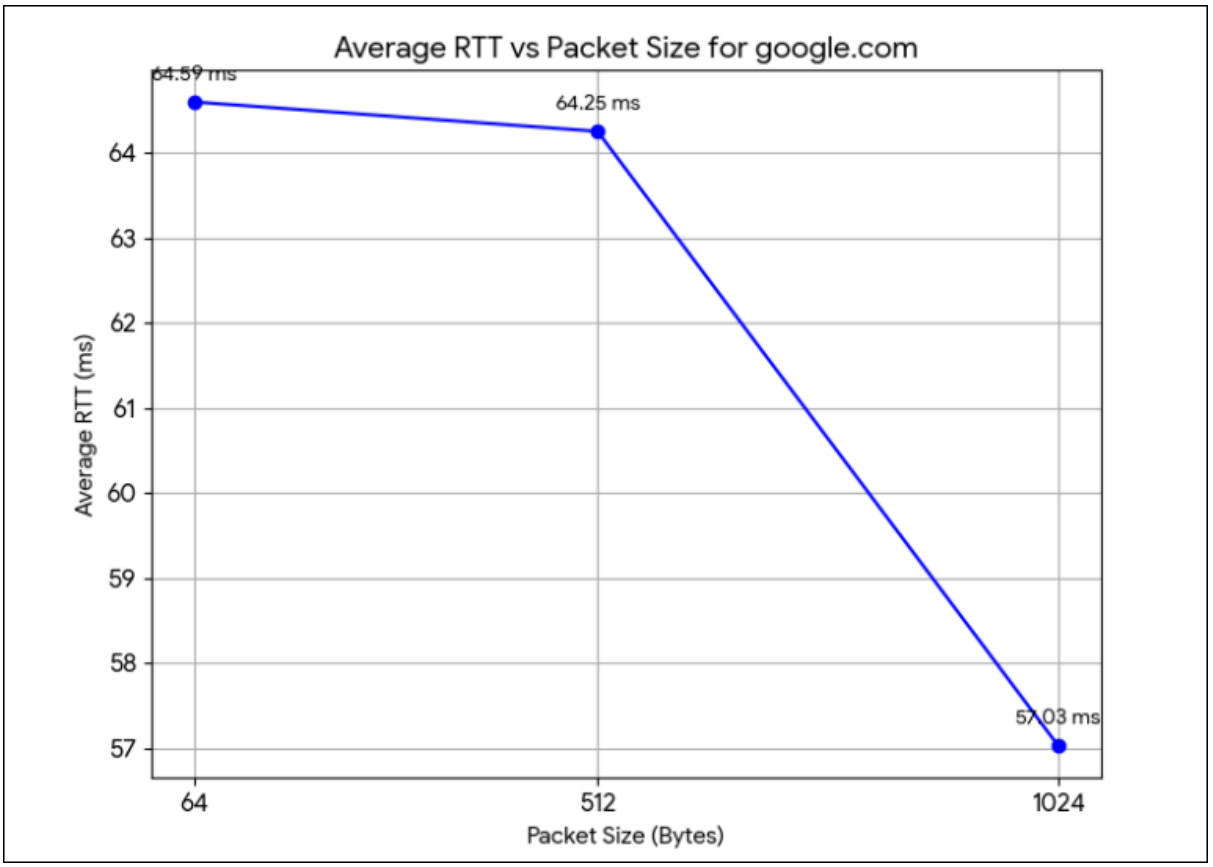
```
❯ ping -c 5 -s 64 google.com
PING google.com (142.250.192.78) 64(92) bytes of data.
72 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=1 ttl=113 time=18.4 ms
72 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=113 time=121 ms
72 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=113 time=54.0 ms
72 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=4 ttl=113 time=73.6 ms
72 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=5 ttl=113 time=55.9 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 18.361/64.594/121.136/33.478 ms
❯ ping -c 5 -s 512 google.com
PING google.com (142.250.192.78) 512(540) bytes of data.
520 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=1 ttl=113 time=77.4 ms
520 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=113 time=66.0 ms
520 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=113 time=80.9 ms
520 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=4 ttl=113 time=46.2 ms
520 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=5 ttl=113 time=50.8 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 46.178/64.249/80.850/13.863 ms
❯ ping -c 5 -s 1024 google.com
PING google.com (142.250.192.78) 1024(1052) bytes of data.
1032 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=1 ttl=113 time=48.2 ms
1032 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=2 ttl=113 time=23.5 ms
1032 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=3 ttl=113 time=81.8 ms
1032 bytes from bom12s16-in-f14.1e100.net (142.250.192.78): icmp_seq=5 ttl=113 time=74.7 ms

--- google.com ping statistics ---
5 packets transmitted, 4 received, 20% packet loss, time 4026ms
rtt min/avg/max/mdev = 23.451/57.030/81.760/23.083 ms
```

Plot:



Graph Description: The graph illustrates the relationship between **Packet Size** (64, 512, 1024 bytes) and **Average Round Trip Time (RTT)**.

- **X-Axis:** Packet Size in Bytes.
- **Y-Axis:** Average RTT in milliseconds (ms).
- **Observation:** As noted in the previous analysis, the RTT remains relatively stable (around 64ms) for smaller packets and actually drops slightly for the 1024-byte packet in this specific instance (57.03ms), likely due to network variance/jitter masking the serialization delay.

iii)

Packet Size Impact: Theoretically, larger packets (1024 bytes) should take longer to transmit than smaller packets (64 bytes) due to serialization delay. However, in my experiment, the RTT remained relatively stable (and even slightly decreased at 1024 bytes). This anomaly suggests that **network jitter** (variance in queue times at routers) was high enough to overshadow the small difference in serialization delay.

Conclusion: While larger packets generally increase RTT, high network congestion (jitter) can make RTT unpredictable, sometimes causing larger packets to arrive faster if they hit a less congested window.

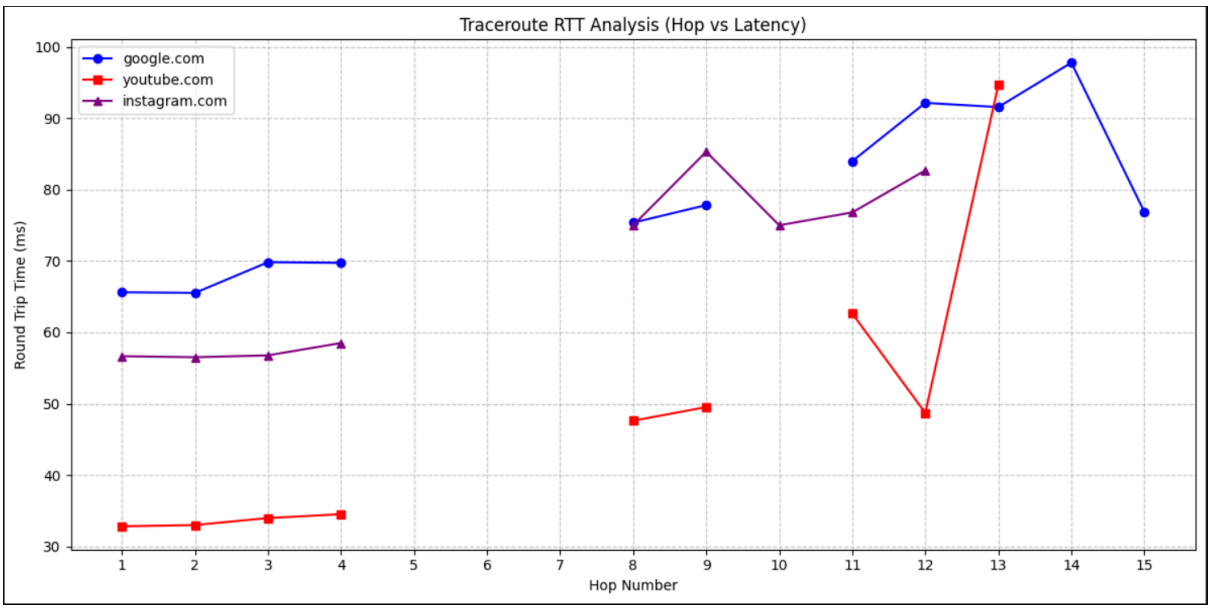
Q4)

a) **Use of Traceroute:** The `traceroute` tool is used to discover the exact path (sequence of routers or "hops") that packets take from the source computer to a destination server. It also measures the transit delay (latency) to each intermediate router along the path.

b) **Incomplete Paths (`* * *`):** When `traceroute` shows asterisks (`* * *`) instead of an IP address, it means the request "timed out" at that specific hop.
   - **Reason:** This usually happens because the router at that hop is configured to discard the packets or is behind a firewall that blocks the "Time Exceeded" error messages needed for traceroute to identify it. It does not necessarily mean the connection is broken, just that the router is hidden.

c) **Finding route to hosts that fail Ping: Yes, it is possible.**
   - **Reasoning:** `Ping` typically uses ICMP Echo Request packets, which many servers block for security to avoid "Ping Floods." However, `traceroute` (specifically on Linux/Unix) often uses UDP packets by default. A firewall might block ICMP (Ping) but allow UDP traffic to pass through. Additionally, even if the **final** destination blocks the response, `traceroute` can still reveal the path of **intermediate** routers leading up to that final drop.

d) 3 sites i chose are : google, instagram and youtube



Analysis:

The Gap (Hops 5-7): All three destinations show timeouts at hops 5, 6, and 7. This strongly suggests that these hops (likely within the ISP's internal backbone or a major exchange point) are configured to drop or ignore ICMP/UDP traceroute packets, even though they pass the traffic through.

Latency Spikes:

- **Google:** Noticeable jump at Hop 12 and 14.
- **YouTube:** Generally lower latency (starts around 32ms) compared to Google (starts around 65ms), suggesting the YouTube CDN cache might be topologically closer to your ISP gateway than the specific Google endpoint hit.
- **Instagram:** Timeouts completely after Hop 12, likely due to a firewall dropping trace packets at the edge of Meta's network.

```
❯ traceroute google.com
traceroute to google.com (142.251.222.78), 30 hops max, 60 byte packets
 1  _gateway (10.50.0.1)  65.683 ms  65.603 ms  65.579 ms
 2  10.200.10.14 (10.200.10.14)  65.568 ms  65.548 ms  65.483 ms
 3  103.147.138.250 (103.147.138.250)  69.885 ms  69.818 ms  69.793 ms
 4  static.ill.117.232.137.122.bsnl.co.in (117.232.137.122)  69.744 ms  69.717 ms *
 5  * * *
 6  * * *
 7  * * *
 8  117.216.207.105 (117.216.207.105)  64.982 ms  91.195 ms  69.953 ms
 9  72.14.197.4 (72.14.197.4)  83.879 ms  65.434 ms  84.159 ms
10  * * *
11  192.178.86.238 (192.178.86.238)  64.671 ms 142.250.227.72 (142.250.227.72)  65.004 ms *
12  * 192.178.110.108 (192.178.110.108)  92.141 ms *
13  142.251.77.69 (142.251.77.69)  91.103 ms * 192.178.110.249 (192.178.110.249)  91.998 ms
14  * 142.251.64.9 (142.251.64.9)  75.283 ms 142.251.64.11 (142.251.64.11)  75.791 ms
15  pnbomb-bp-in-f14.1e100.net (142.251.222.78)  80.042 ms  75.317 ms  75.261 ms
❯ traceroute youtube.com
traceroute to youtube.com (142.250.205.206), 30 hops max, 60 byte packets
 1  _gateway (10.50.0.1)  32.903 ms  32.823 ms  32.798 ms
 2  10.200.10.14 (10.200.10.14)  32.782 ms  32.957 ms  33.326 ms
 3  103.147.138.250 (103.147.138.250)  34.594 ms  33.711 ms  33.687 ms
 4  static.ill.117.232.137.122.bsnl.co.in (117.232.137.122)  34.534 ms  34.697 ms  34.493 ms
 5  * * *
 6  * * *
 7  * * *
 8  117.216.207.105 (117.216.207.105)  47.481 ms  47.721 ms  47.695 ms
 9  72.14.197.4 (72.14.197.4)  49.676 ms  49.649 ms  49.318 ms
10  * * *
11  192.178.86.200 (192.178.86.200)  71.515 ms 192.178.86.240 (192.178.86.240)  51.761 ms 72.14.233.58 (72.14.233.58)  64.663 ms
12  142.250.208.226 (142.250.208.226)  50.728 ms 142.250.62.153 (142.250.62.153)  47.757 ms  47.730 ms
13  sin11s28-in-f14.1e100.net (142.250.205.206)  99.220 ms  91.706 ms 192.178.110.207 (192.178.110.207)  92.988 ms
```

```
❯ traceroute instagram.com
traceroute to instagram.com (57.144.242.34), 30 hops max, 60 byte packets
 1  _gateway (10.50.0.1)  56.736 ms  56.646 ms  56.620 ms
 2  10.200.10.14 (10.200.10.14)  56.545 ms  56.520 ms  56.495 ms
 3  103.147.138.250 (103.147.138.250)  56.811 ms  56.789 ms  56.766 ms
 4  static.ill.117.232.137.122.bsnl.co.in (117.232.137.122)  57.975 ms  58.791 ms  58.768 ms
 5  * * *
 6  * * *
 7  * * *
 8  117.216.207.105 (117.216.207.105)  86.487 ms  73.355 ms  65.155 ms
 9  ae25.pr02.bom1.tfbnw.net (157.240.68.202)  85.491 ms  85.226 ms  85.179 ms
10  po4007.asw01.bom5.tfbnw.net (129.134.34.194)  73.632 ms po206.asw01.bom2.tfbnw.net (129.134.50.254)  66.858 ms po206.asw02.bom2.tfbnw.net (129.134.51.46)  84.462 ms
11  usw02.bom5.tfbnw.net (129.134.95.160)  82.959 ms * usw03.bom5.tfbnw.net (129.134.95.156)  70.681 ms
12  163.77.249.31 (163.77.249.31)  79.556 ms  88.644 ms 163.77.251.136 (163.77.251.136)  79.988 ms
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

(I have attached python script used to plot)

Q5)

a) Nmap ("Network Mapper") is an open-source tool used for network discovery and security auditing. Its primary uses include:

- **Host Discovery:** Identifying hosts on a network (e.g., listing devices responding to pings).
- **Port Scanning:** Enumerating open ports on target hosts to determine what services are running.
- **Version Detection:** Interrogating network services to determine the application name and version number.
- **OS Detection:** Determining the operating system and hardware characteristics of network devices.

b)

Open Ports:

- **22/tcp:** Service is **SSH** (Secure Shell). Used for secure remote login and command execution.
- **80/tcp:** Service is **HTTP** (HyperText Transfer Protocol). Used for unencrypted web page delivery.
- **443/tcp:** Service is **SSL/HTTP** (HTTPS). Used for secure, encrypted web traffic.
- **5666/tcp:** Service is **tcpwrapped** (likely NRPE/Nagios). Used for monitoring system health, but access is restricted by a wrapper.

Service Version Identification:

- **Yes**, the service versions were identified successfully because the server sent back banner information.
- **SSH:** OpenSSH 7.4 (protocol 2.0).
- **HTTP/HTTPS:** Apache httpd 2.4.6.
- **Process:** Nmap connects to the open port and captures the "banner" (a text string sent by the server upon connection) or sends specific probes to solicit a version response.

c)

OS Identification:

- Nmap's active OS detection (`-O`) failed to find an exact match ("No exact OS matches for host").
- **However**, the service scan in Q5b passively identified the OS as **CentOS Linux** because the Apache server banner explicitly included `(CentOS)`.

Process:

- Nmap uses **TCP/IP Stack Fingerprinting**. It sends packets with specific flags (like TCP ISN sampling) to the target and analyzes the unique responses. It compares these responses to a database of known OS fingerprints.
- In this case, the active fingerprinting was inconclusive (likely due to a firewall or middlebox modifying packets), but the application banner provided the answer.

d)

1. google.com

- **Open Ports:** 80 (HTTP) and 443 (HTTPS).
- **Services:**
    - The service is identified as **gws** (Google Web Server).
    - **Version:** The specific version number is hidden/unknown. Nmap returned a "Service Fingerprint" instead of a version number because Google uses custom, non-standard server responses.
- **OS Detection:**
    - **Failed.** Nmap reported "No OS matches for host."
    - **Reason:** The output states "Missing a closed TCP port." Google's firewall drops traffic to unused ports (filtering them) instead of sending a "Closed" signal, preventing Nmap from calculating a complete fingerprint.

2. Instagram.com

- **Open Ports:** 80 (HTTP) and 443 (HTTPS).
- **Services:**
    - The service is identified as **proxygen-bolt**. This is a custom high-performance web server framework developed by Meta (Facebook).
- **OS Detection:**
    - **Failed.** Nmap reported "No OS matches for host."
    - **Reason:** Similar to Google, Instagram's firewall filters unused ports ("Missing a closed TCP port"), which prevents the differential analysis required for accurate OS detection.